Site Modeling with LiDAR and Physical Fabrication

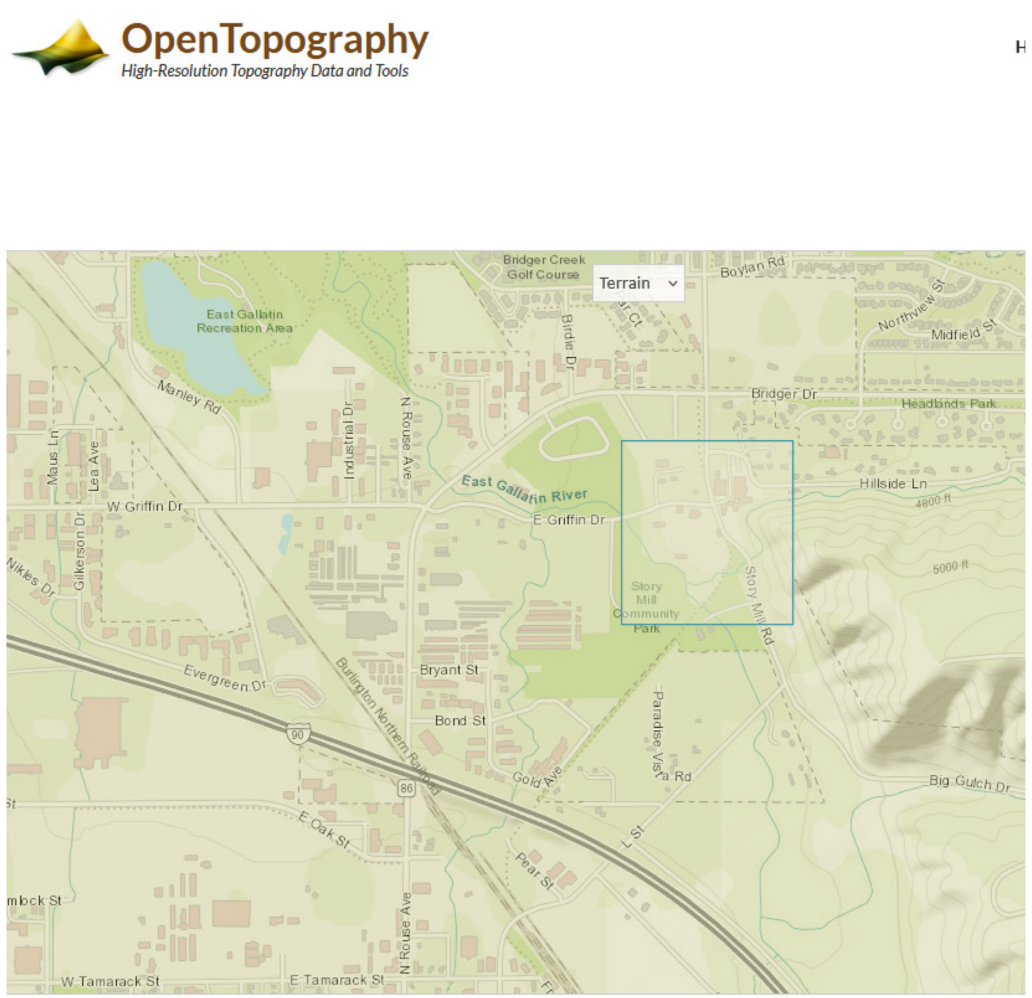
Arch565_Assignment 1

by Alex Tupper

1. Site Selection and Scanning

OpenTopography, Montana State Library

There are a ton of different places to get data from. I think the Montana State Library worked well for the grasshopper scripts, but I pulled the point clouds from OpenTopography because I was able to get a smaller area for use with CloudCompare. I haven't used sources like these before, OpenStreetMaps is my default for anything related to sites, but it worked quite similarly.

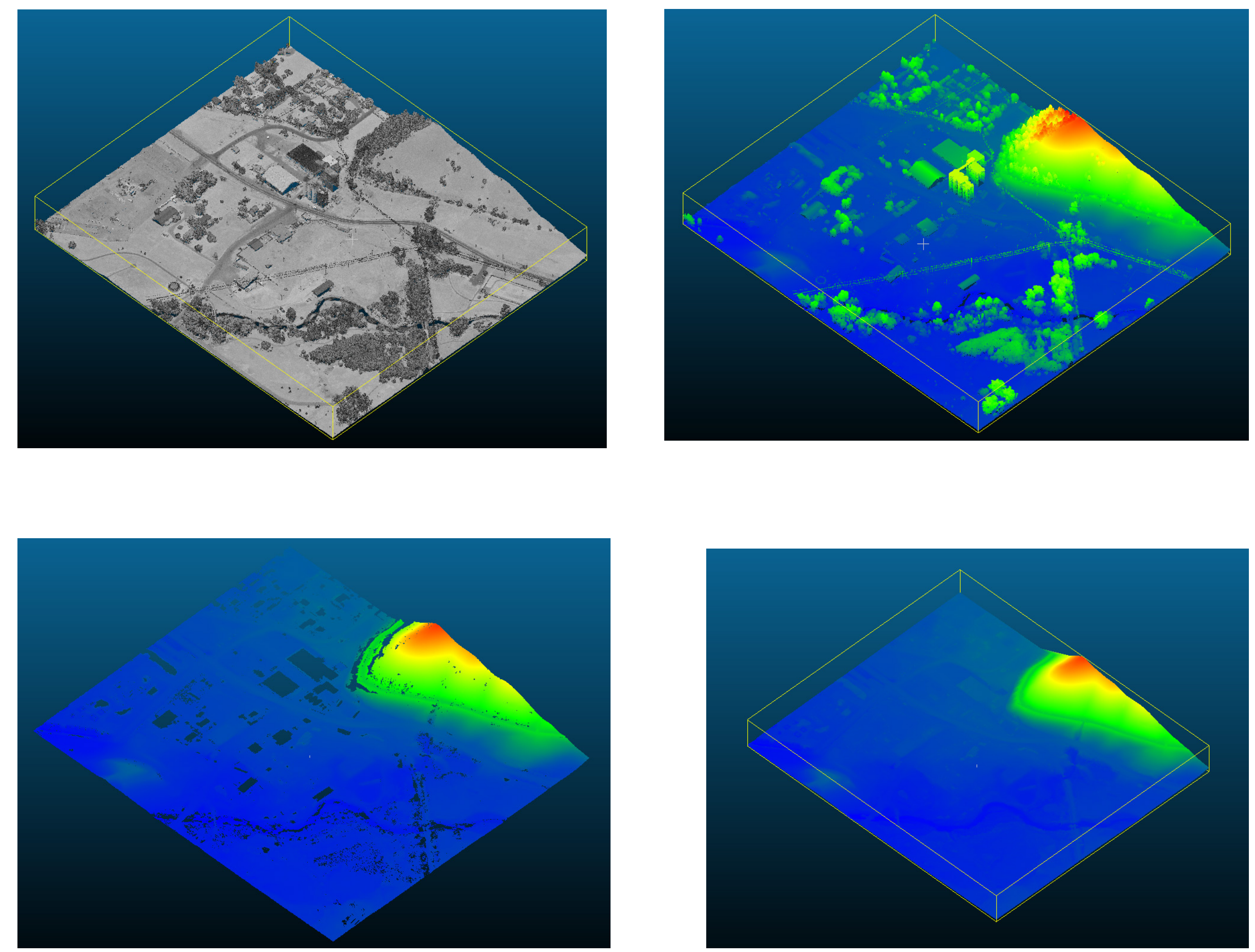


Site on OpenTopography

2. Data Processing

CloudCompare

Using CloudCompare was pretty fun, getting to see how this type of processing works tickled the inquisitive part of my brain. Last semester I used QGIS, got .osm data, exported topolines, then manually moved them up and down for this exact site. This was much easier to generate a mesh with.

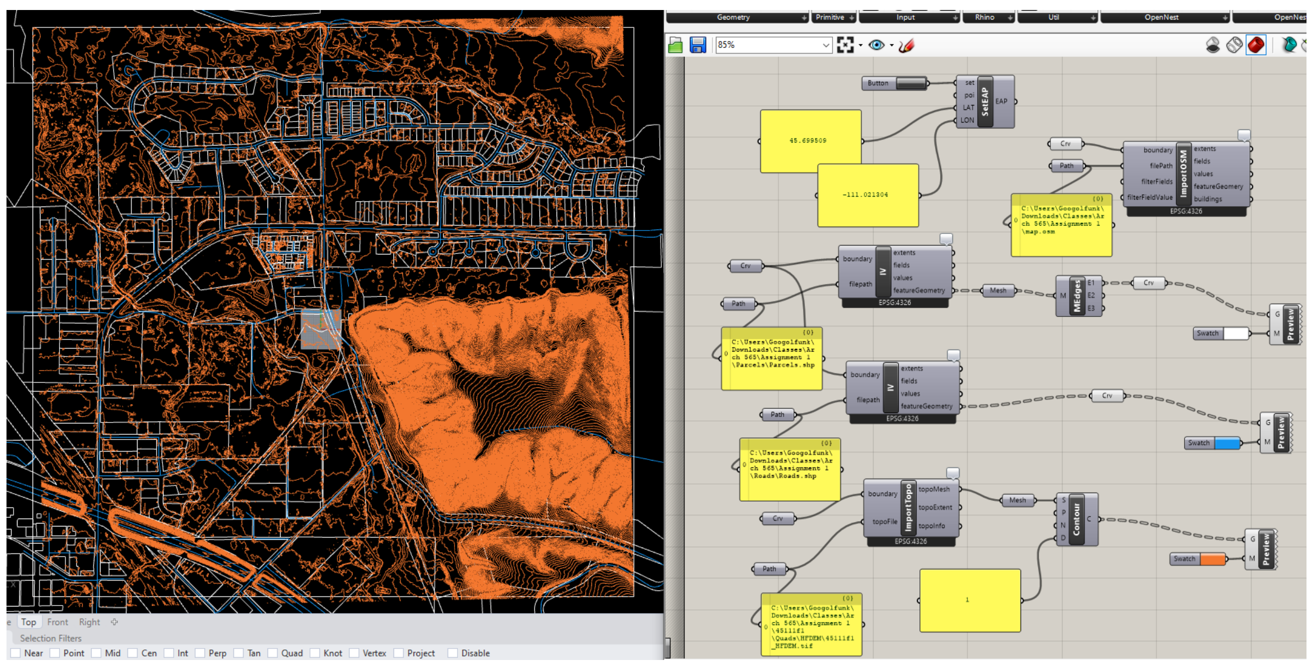


Cloudcompare: Two CSF passes plus mesh

2. Digital Modeling

Heron

This is my Heron script based on the tutorial. It contains parcels, roads, and topography. Again, I manually lined up this information last semester with this exact site and this took much less time. This was an alternative to my CloudCompare mesh, and I think this works better for 2-D information. I could reasonably generate a mesh from the topography lines, but the LiDAR'd mesh from CloudCompare is likely more accurate.

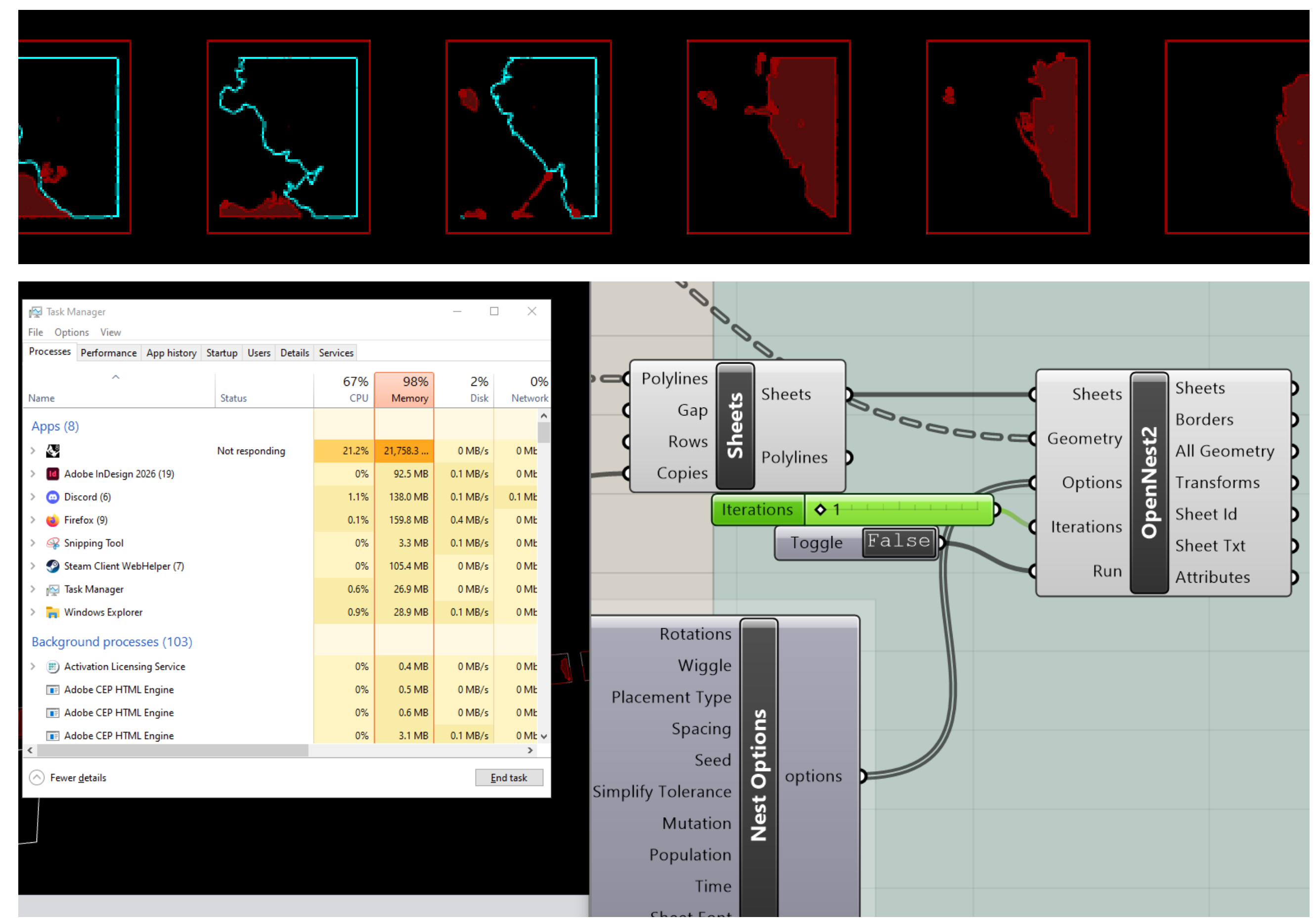


Heron based Grasshopper Script for pulling mapping data

2. Digital Modeling (again)

OpenNest

This is the script uploaded to help us with laying out our sheets and is using my mesh from CloudCompare. I did not make changes to it so I have ommitted it from the reflection. I attempted to get OpenNest to run, but ran into some memory issues. I think I was doing something wrong (I probably just don't know how to use the plugin) but even one interation of nesting ate all my memory.



OpenNest eating my memory and crashing